

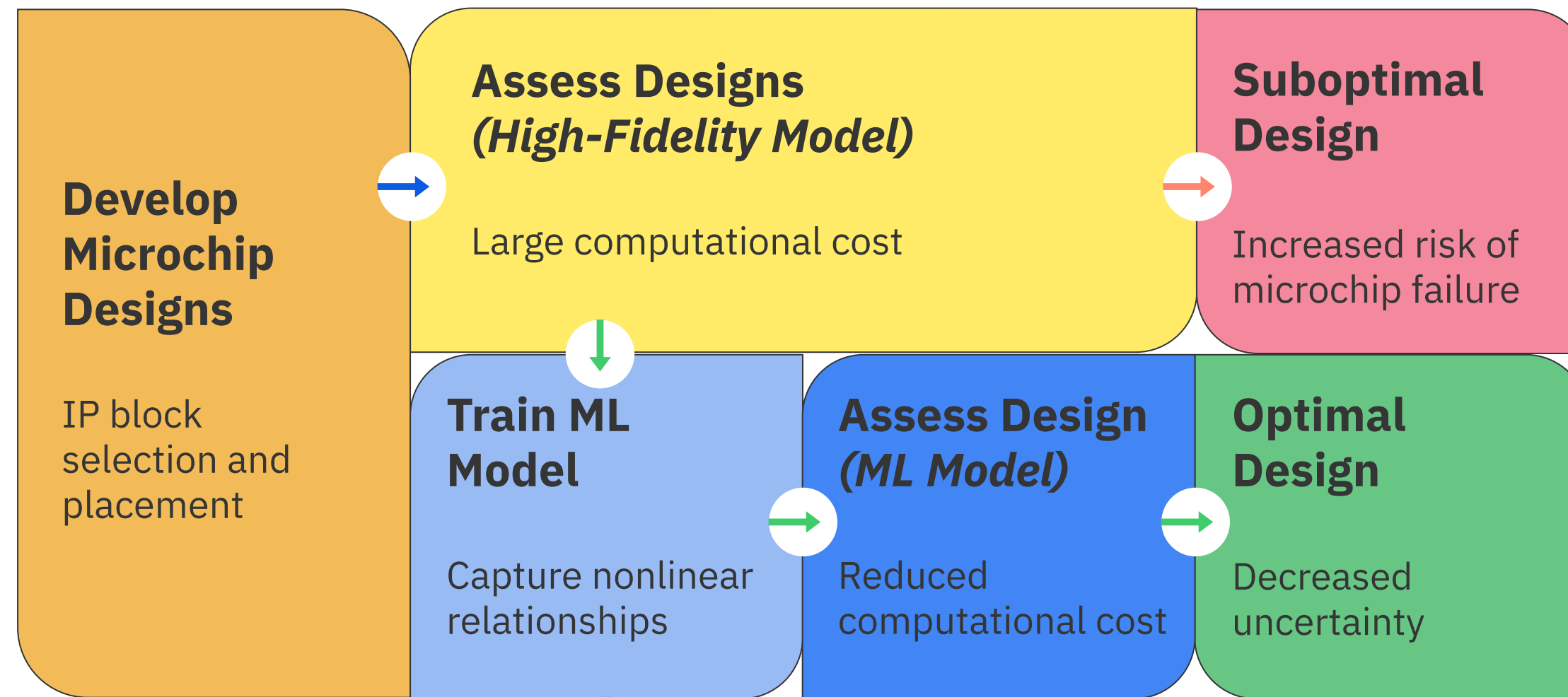
***Surrogate Optimization  
Modeling Temperature and  
Stress in Microchips***

**Maggie Poulsen & Lauren Wylie**

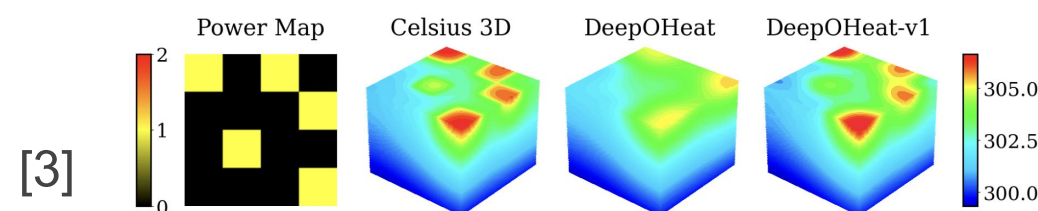
2026 FDS Emerge

# Project Overview

Center for Advanced Electronics through Machine Learning (CAEML)

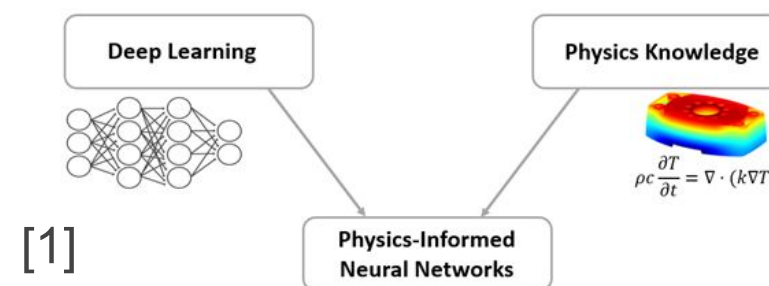


**Spring Semester: DeepOHeat**  
(multi-network operator learning)



**FDS Emerge: PINNs**

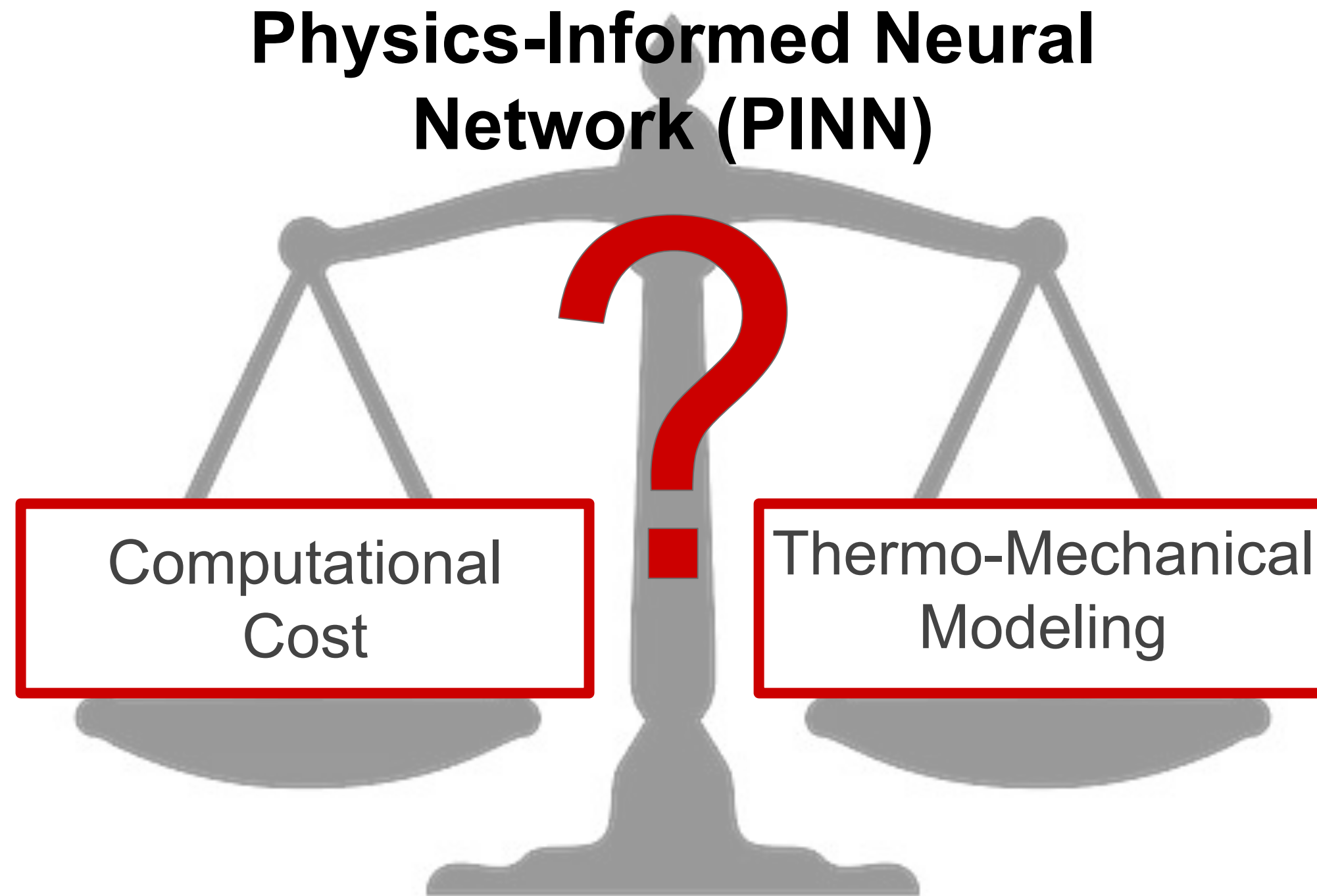
(single-network physics + data learning)



# Project Overview

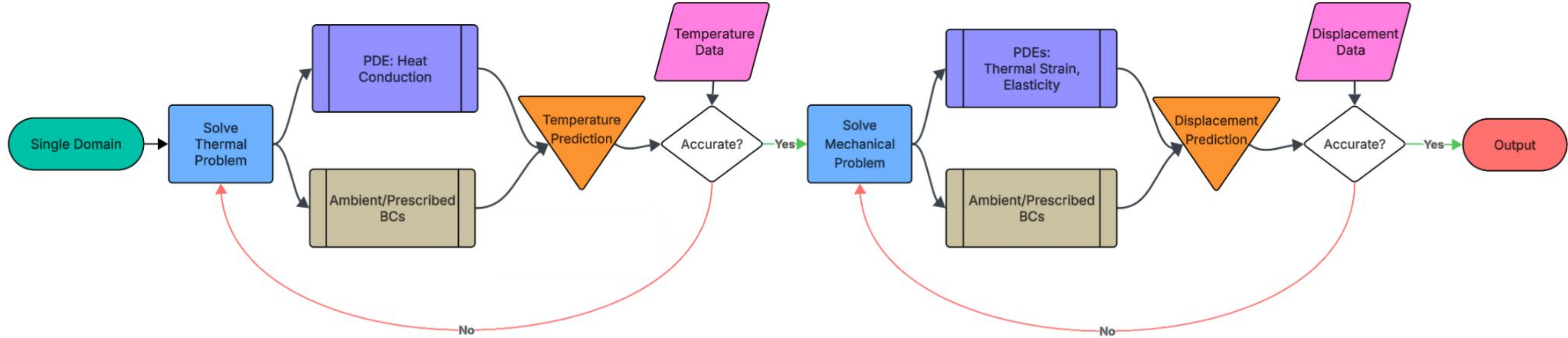
Question of Interest

**Physics-Informed Neural  
Network (PINN)**



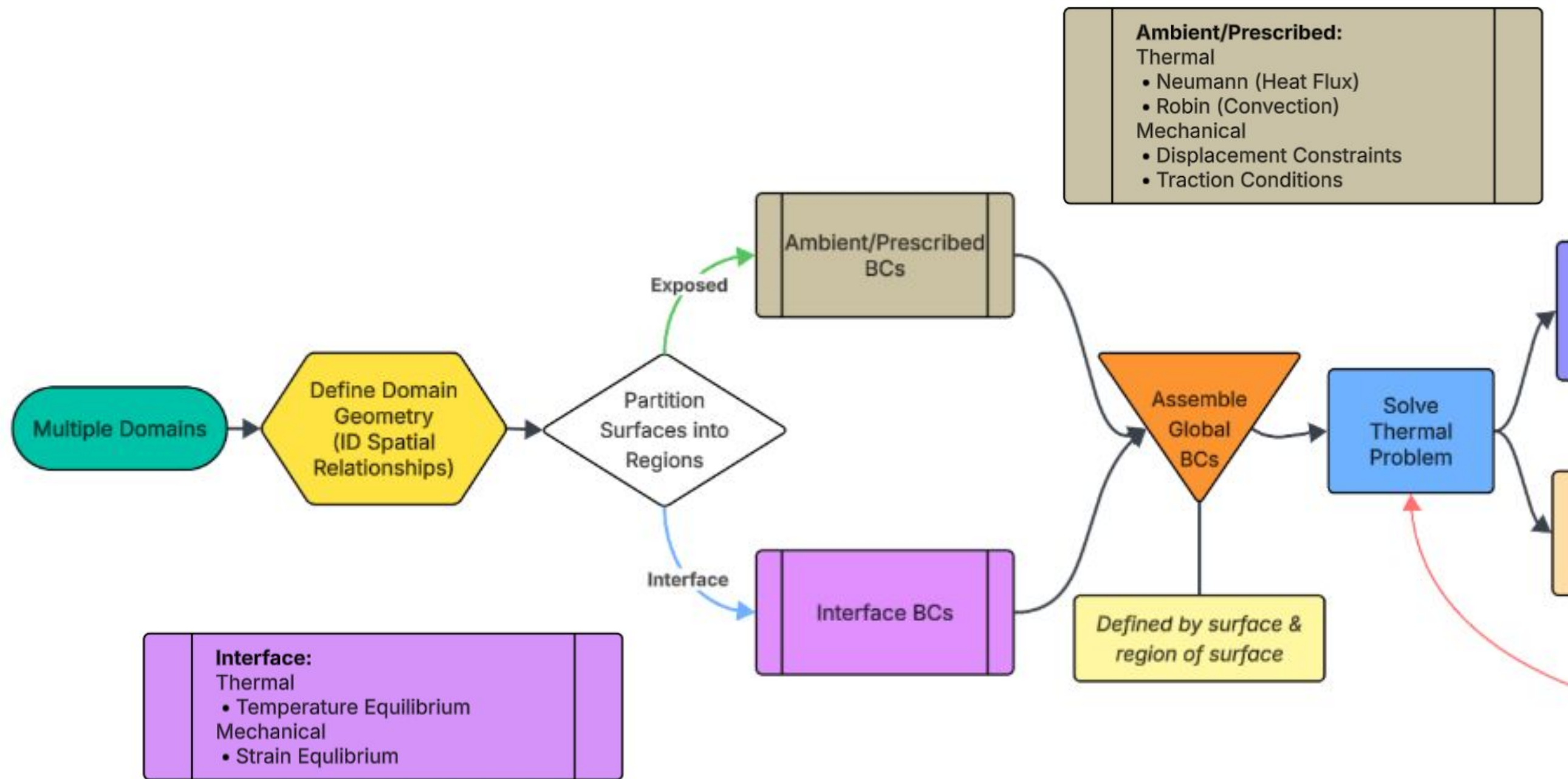
# Single Plate Workflow

## Homogeneous Domain

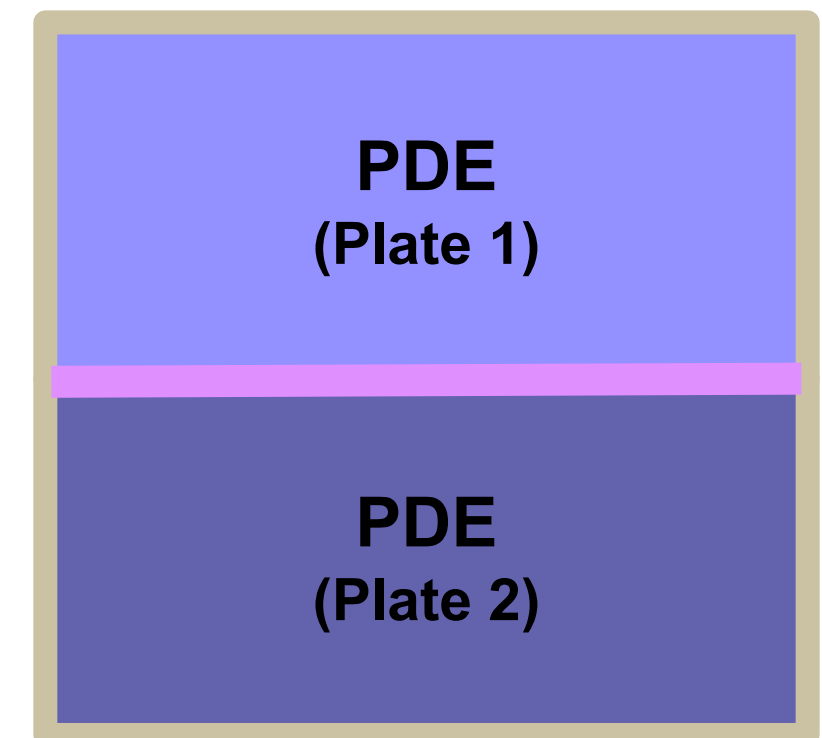


# 2D Assembly (2+ Plates) Workflow

## Equilibrium Boundary Conditions



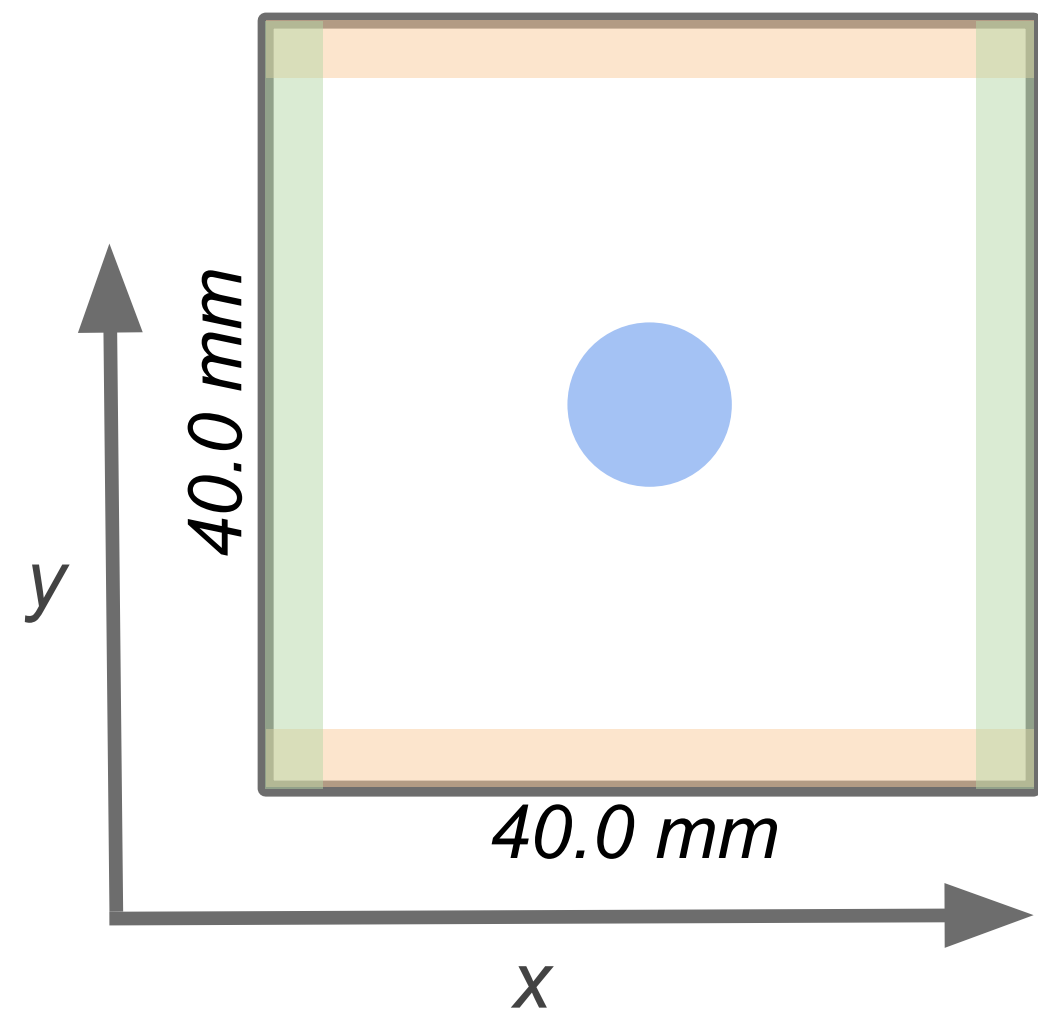
## 2D Assembly



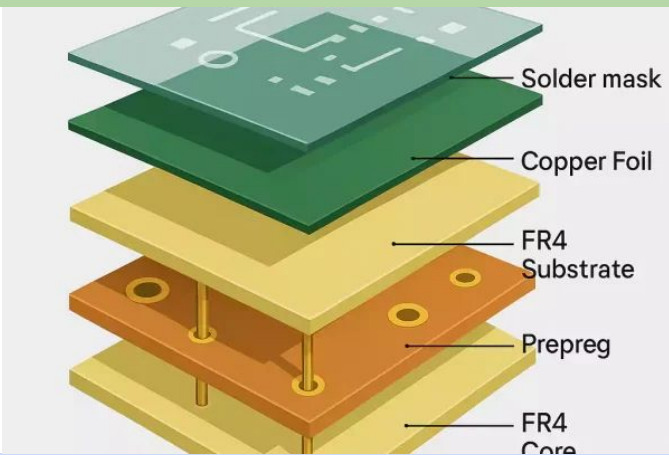


# Design Overview

## Single Plate



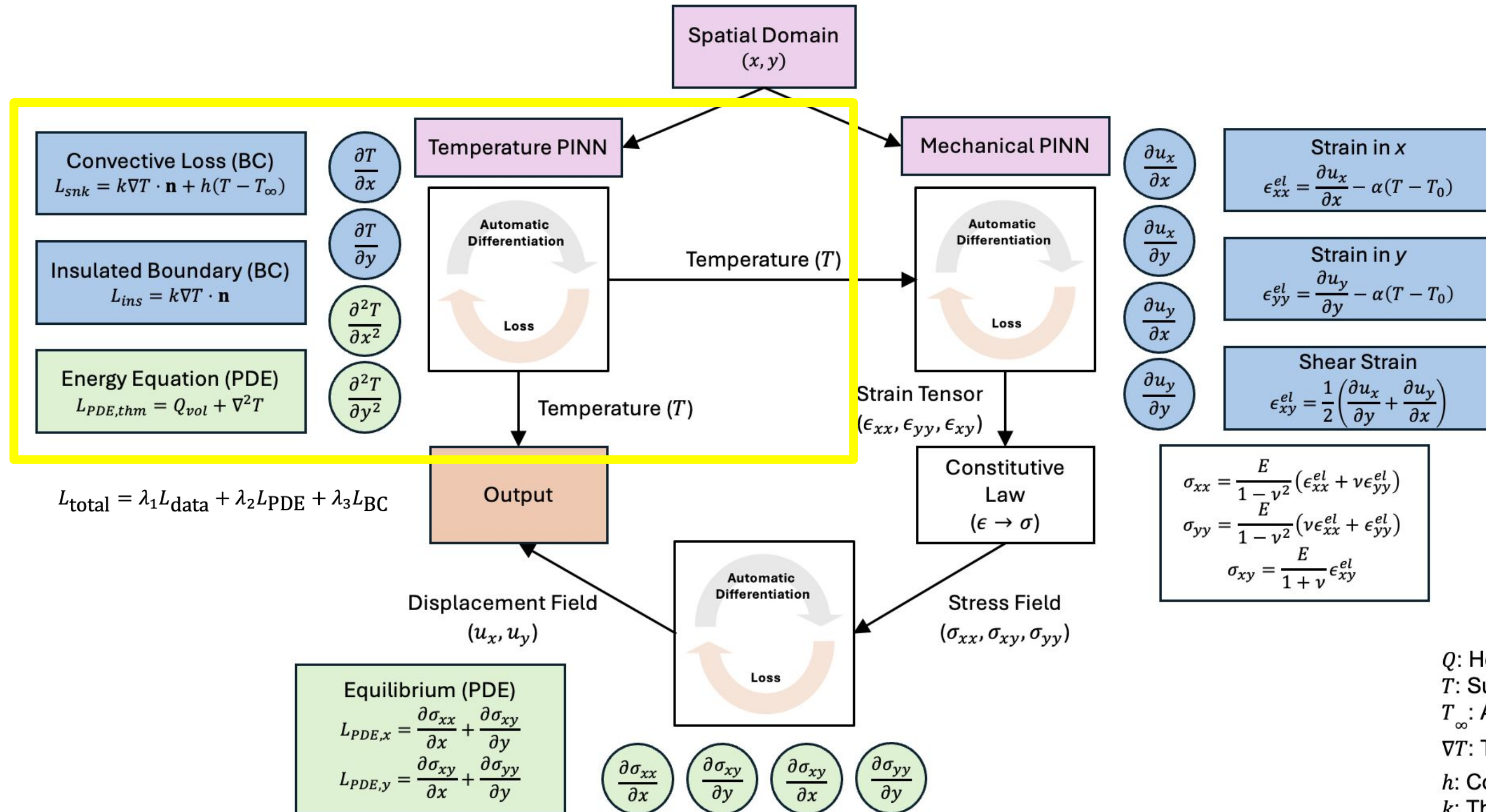
| Surface              | Boundary Condition Type |
|----------------------|-------------------------|
| Bottom ( $y = 0$ )   | Insulated               |
| Top ( $y = L$ )      | Insulated               |
| Left ( $x = 0$ )     | Robin                   |
| Right ( $x = W$ )    | Robin                   |
| <b>Parameters</b>    | <b>Details</b>          |
| Material             | FR-4                    |
| Thermal Conductivity | $k = 0.3$               |
| Natural Convection   | $h = 10$                |
| Heat Source          | Gaussian                |





# Thermomechanical PINN Framework

## Single Plate



$Q$ : Heat flux ( $W/m^2$ )  
 $T$ : Surface Temperature (K)  
 $T_\infty$ : Ambient Temperature (K)  
 $\nabla T$ : Temperature gradient (K/m)  
 $h$ : Convective coefficient ( $W/m^2 K$ )  
 $k$ : Thermal conductivity ( $W/mK$ )  
 $n$ : Outward unit normal to surface

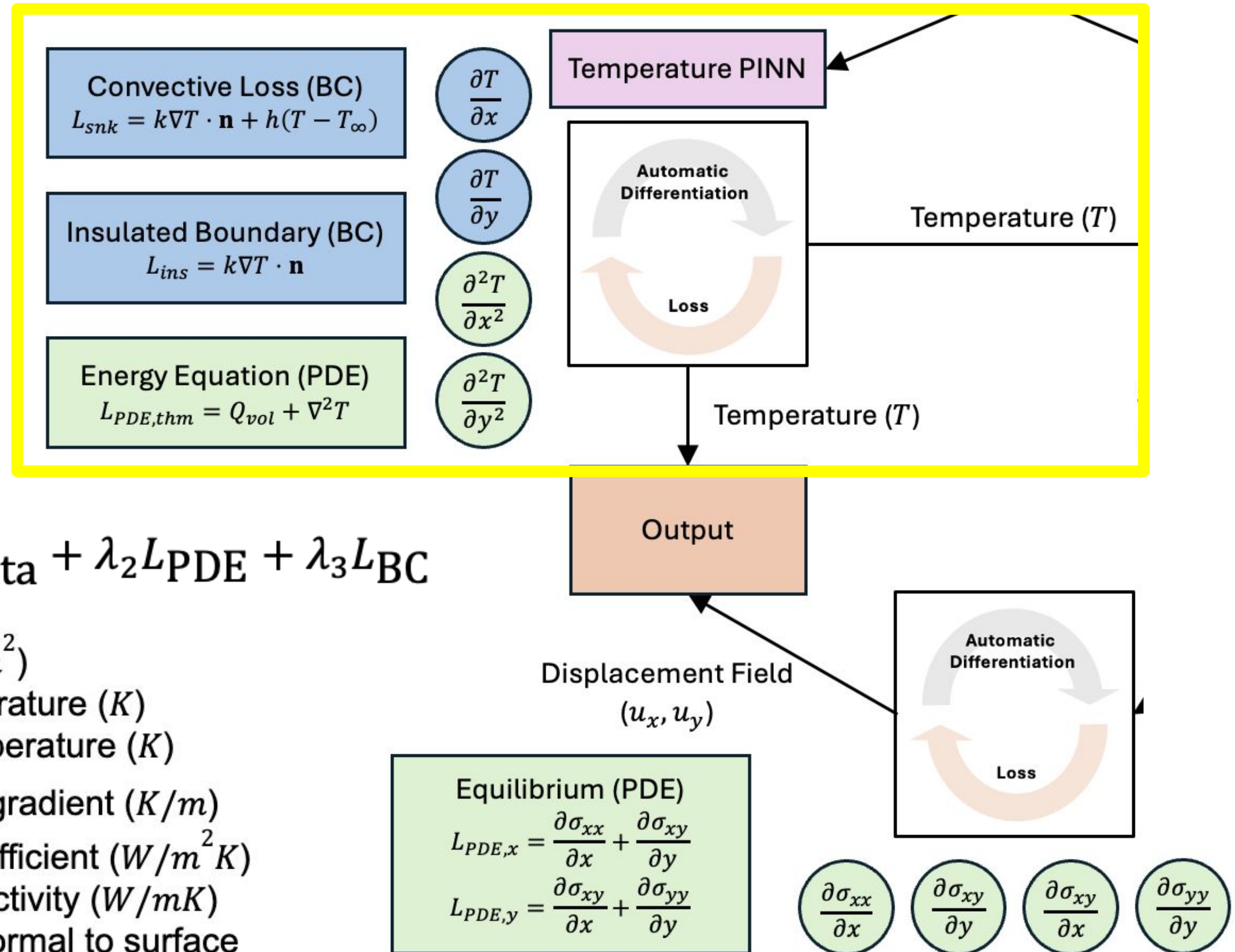
# PINN

## Framework

### Single Plate

Loss = SUM(L-part) / num-parts

L-part = MSE(I-part)\*lambda



$$L_{total} = \lambda_1 L_{data} + \lambda_2 L_{PDE} + \lambda_3 L_{BC}$$

$Q$ : Heat flux ( $W/m^2$ )

$T$ : Surface Temperature ( $K$ )

$T_{\infty}$ : Ambient Temperature ( $K$ )

$\nabla T$ : Temperature gradient ( $K/m$ )

$h$ : Convective coefficient ( $W/m^2 K$ )

$k$ : Thermal conductivity ( $W/mK$ )

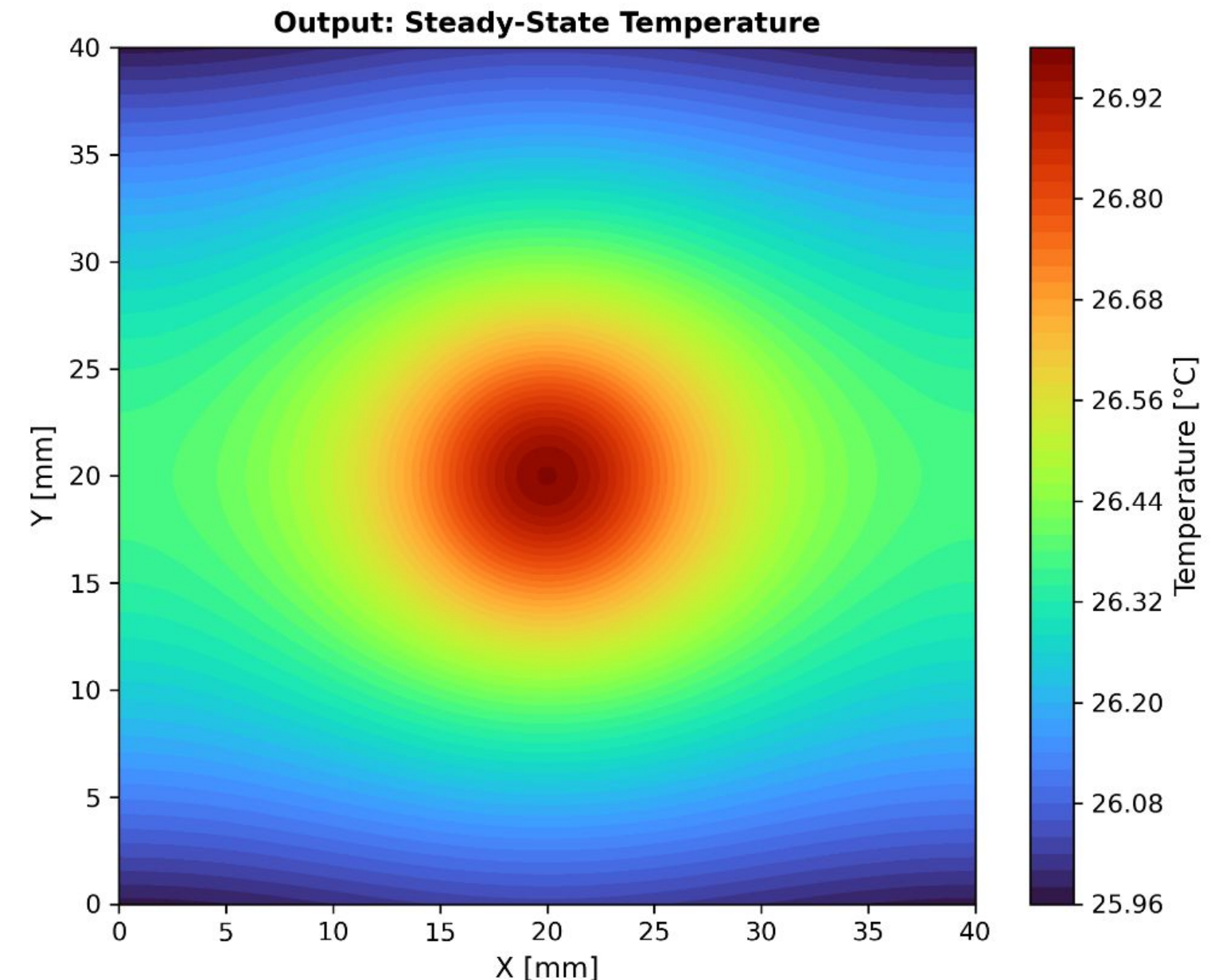
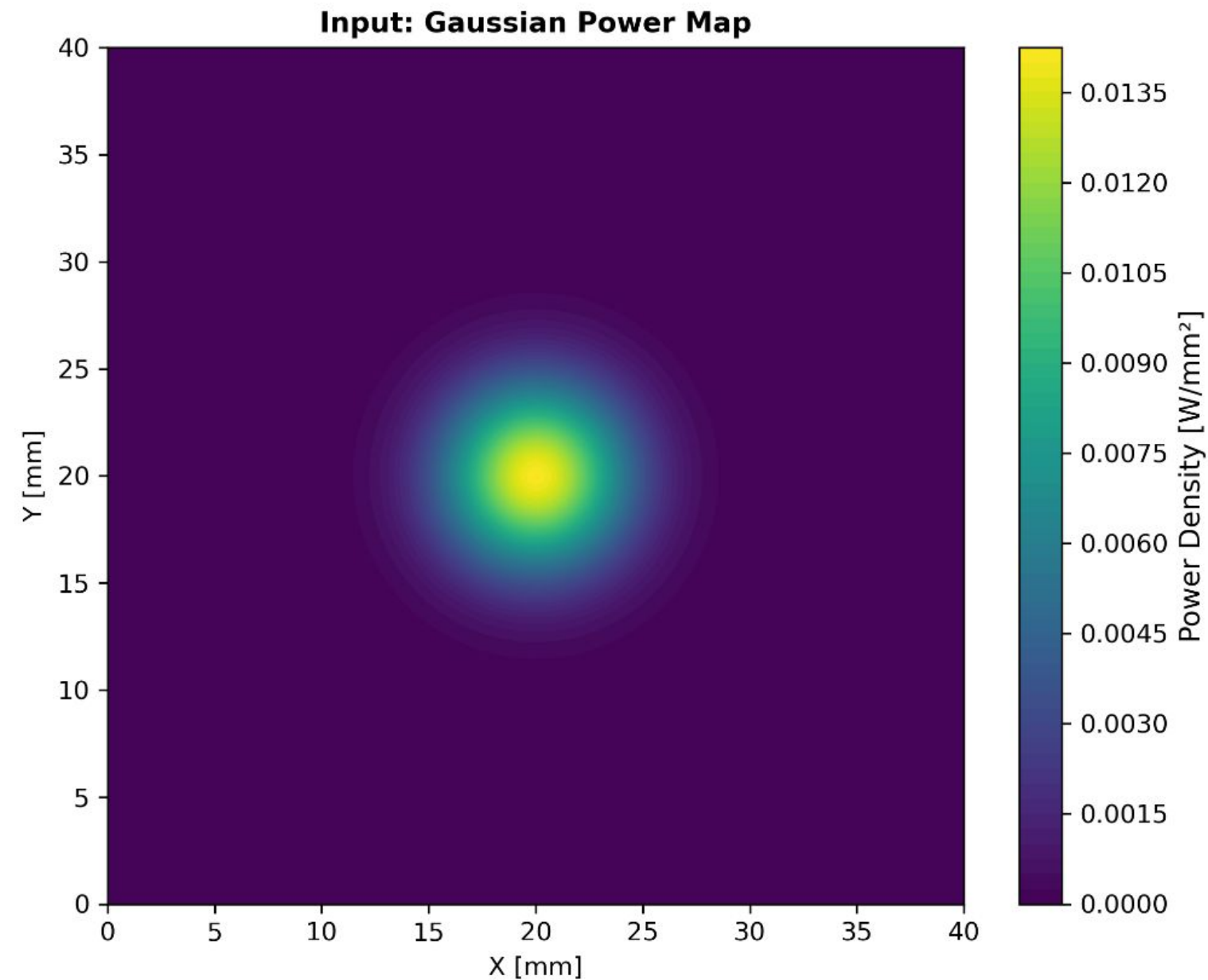
$n$ : Outward unit normal to surface



# Analytical Solution

## Single Sample

**FR-4 Substrate: 2D Thermal Problem**  
**Case: 01 | Source: Center - Low Power & Med Spread**

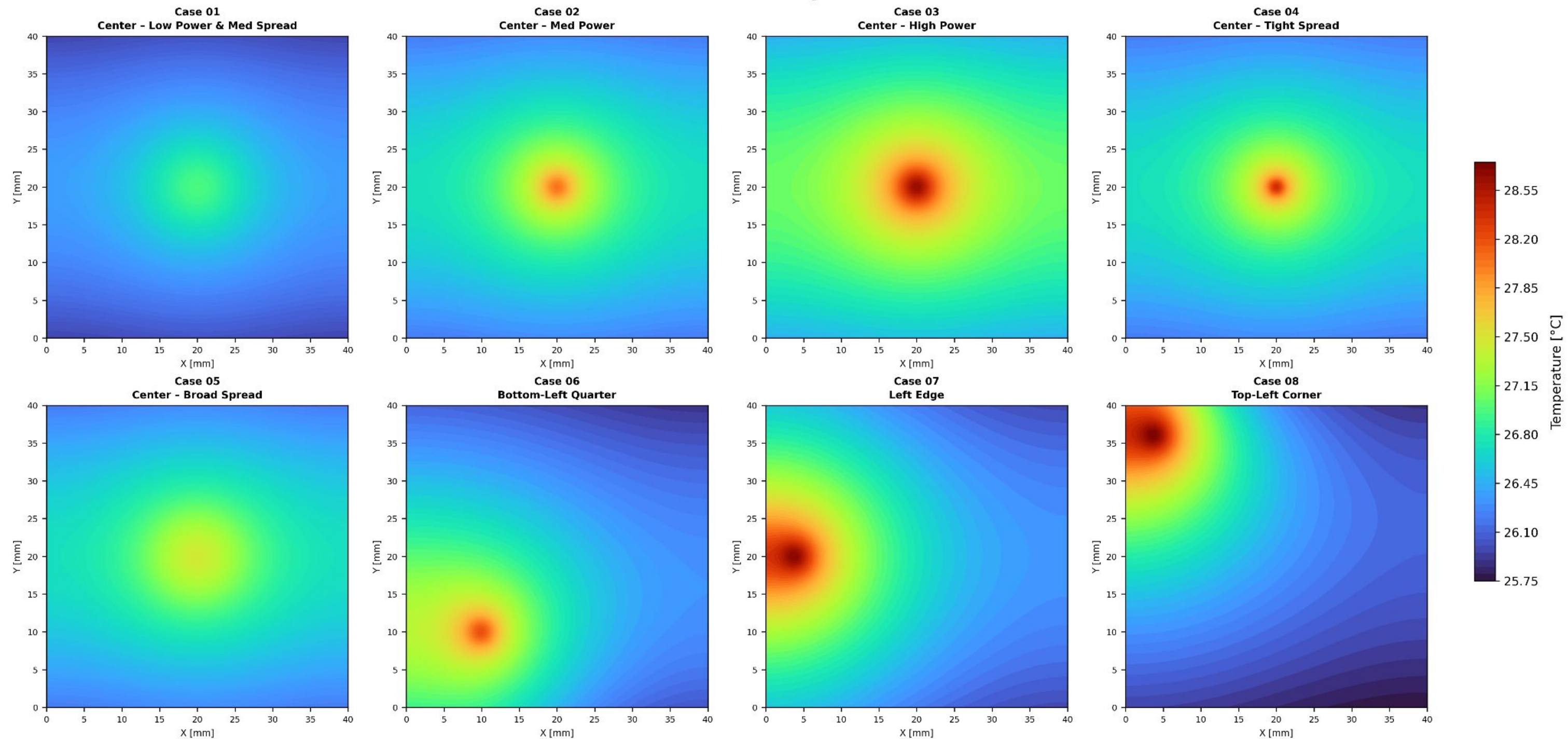


**Boundary States:** Left: Neumann ( $h=0.0$ ,  $q=0.0$ ) | Right: Neumann ( $h=0.0$ ,  $q=0.0$ ) | Bottom: Robin ( $h=10.0$ ,  $q=0.0$ ) | Top: Robin ( $h=10.0$ ,  $q=0.0$ )  
**Source Details:** Power: 0.8 [W] | Location: (20.0, 20.0) | Spread: 3.0 [mm]

# Analytical Solution

## Multiple Samples

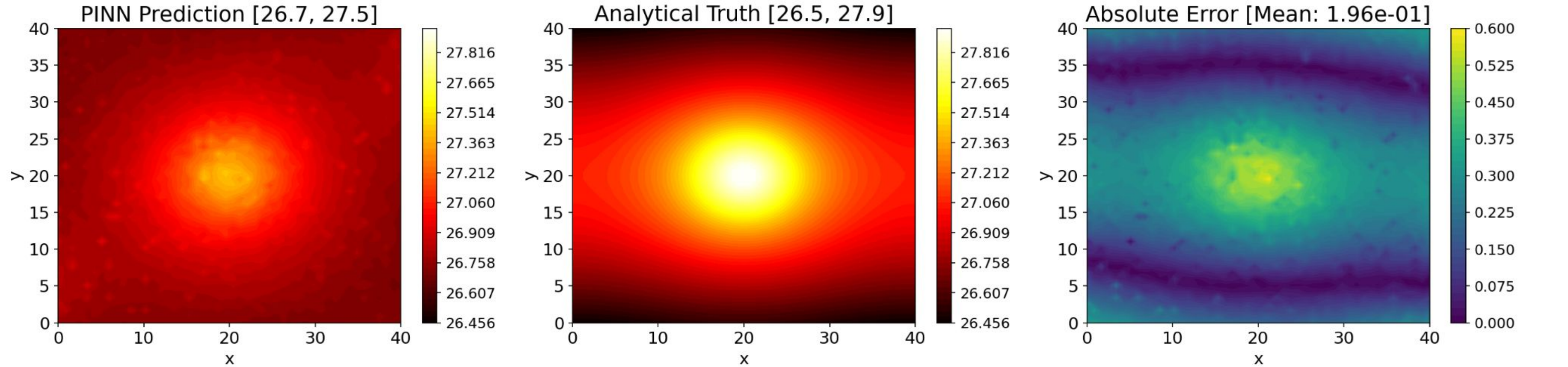
Steady-State Temperature Fields — All Cases  
FR-4 Substrate (Analytical Solution)



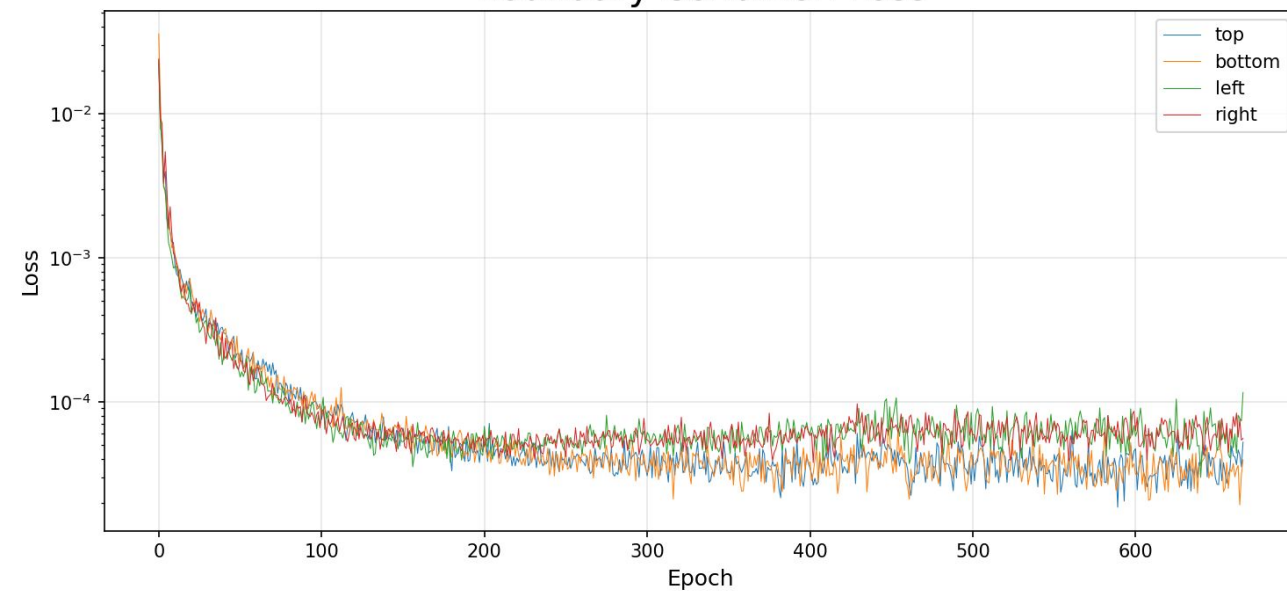


# All BCs Results (Normalized)

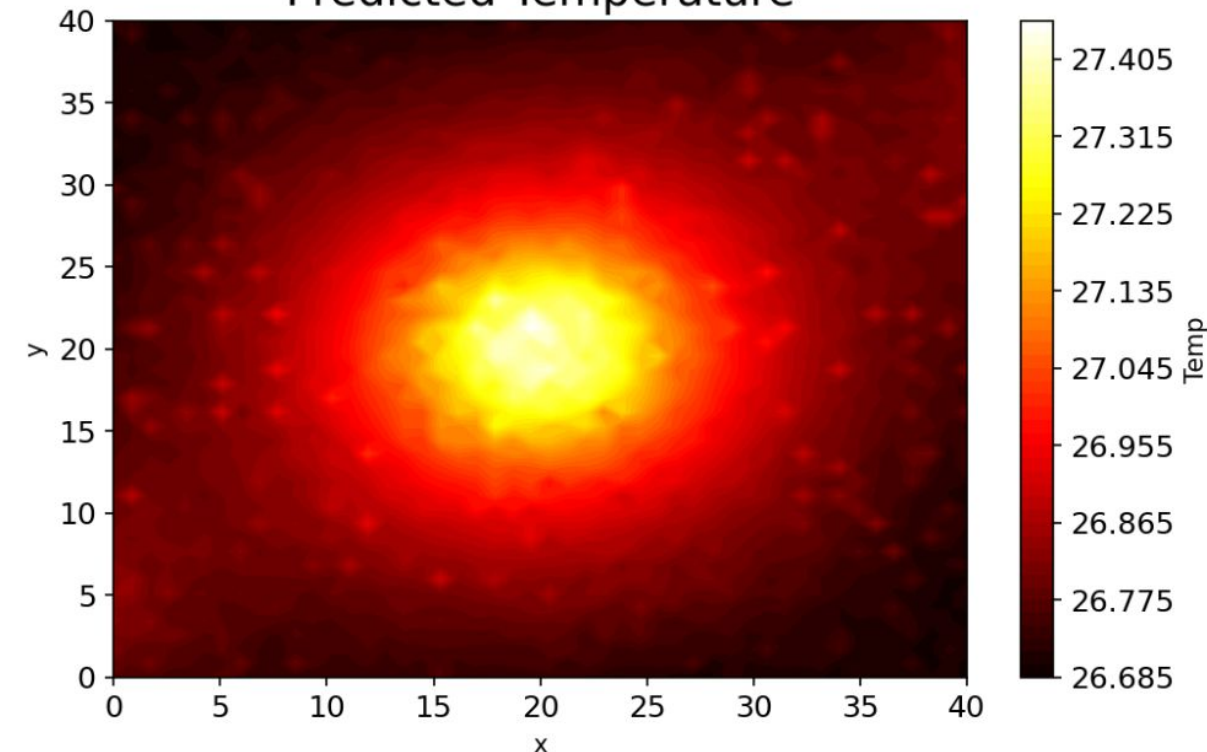
## Gauss Center Power (2000 Epochs)



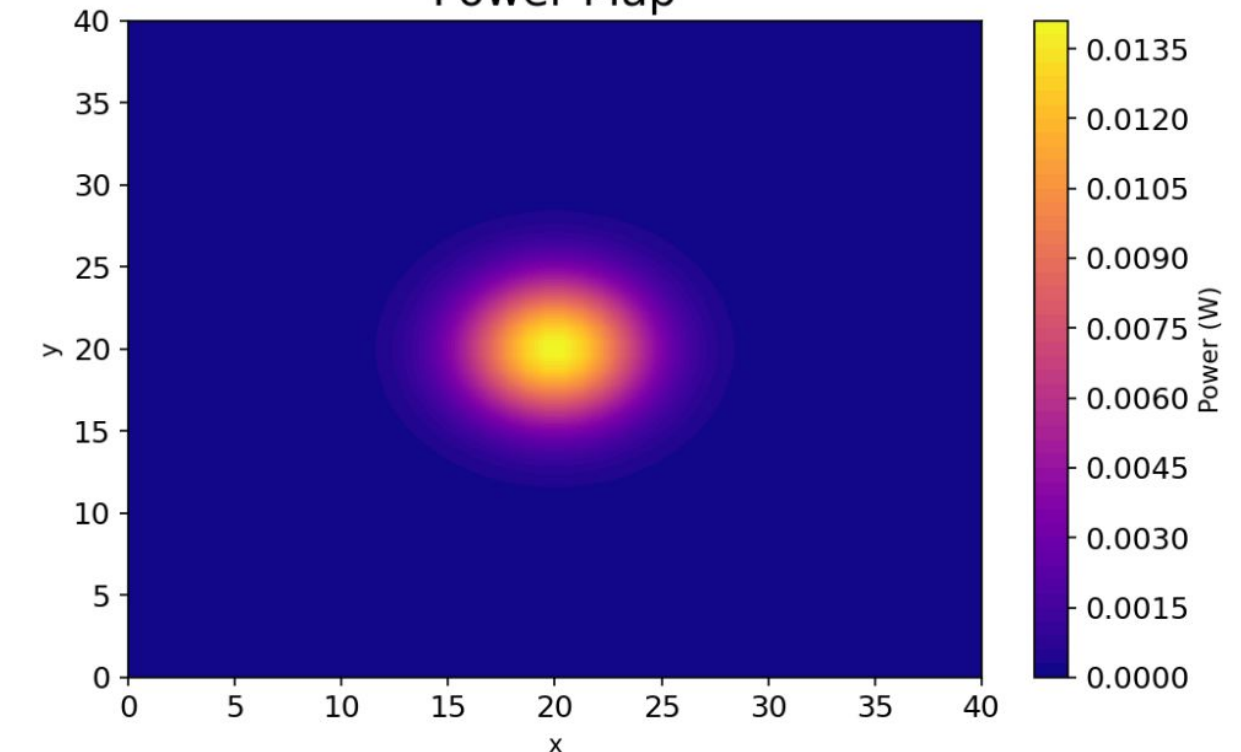
## Boundary Condition Loss



## Predicted Temperature



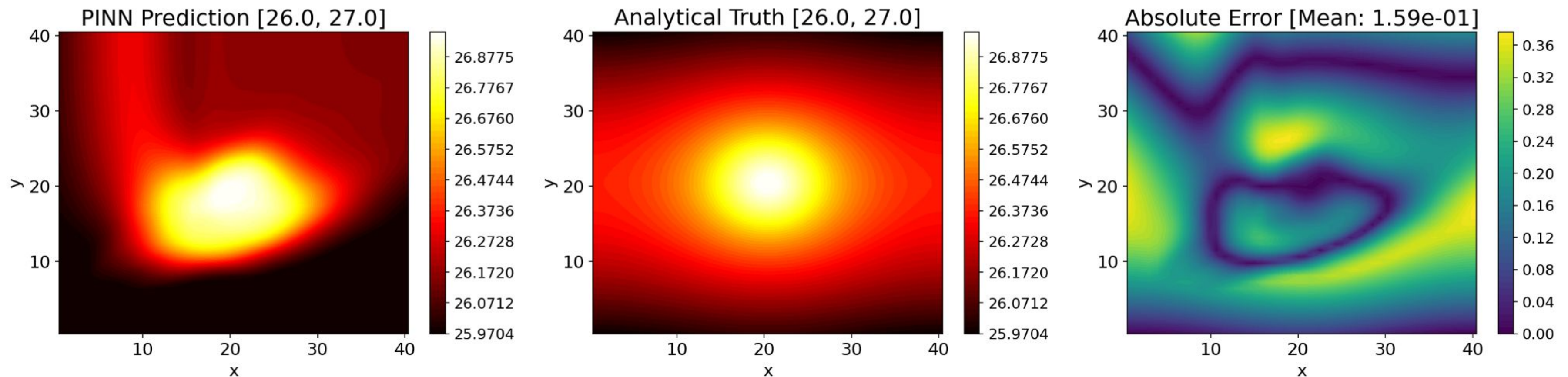
## Power Map





# All BCs Results (Ambient-Min, Convergence)

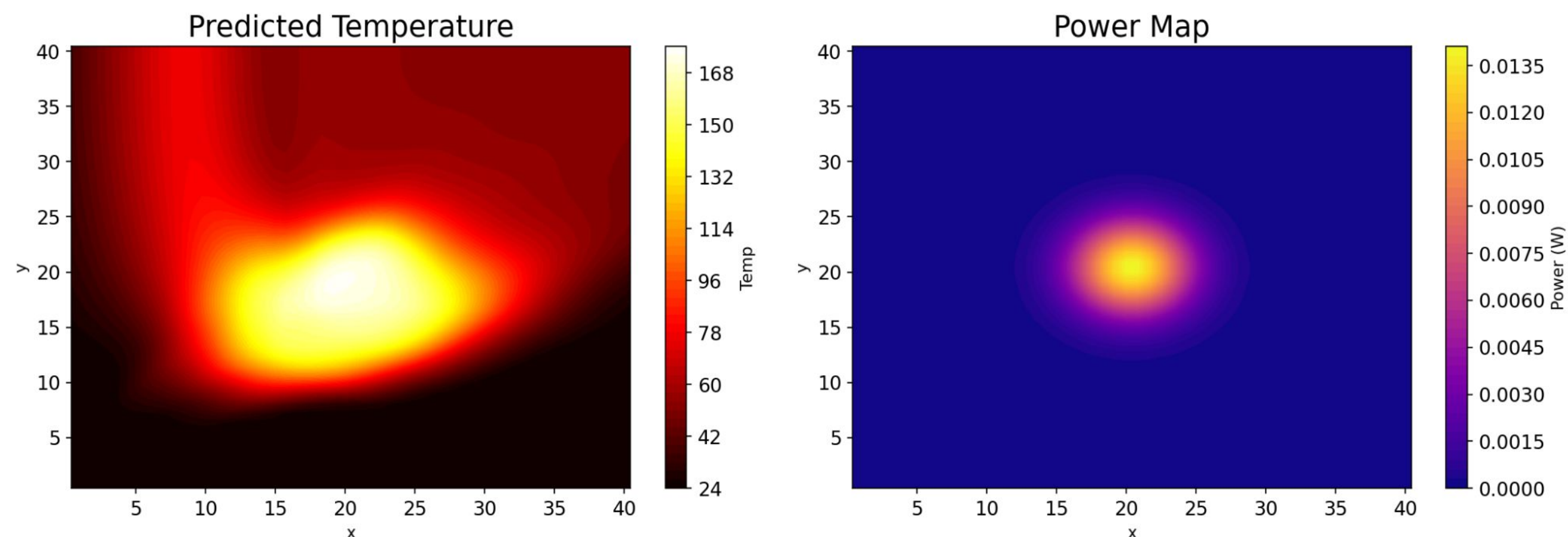
Gauss Power With BCs 20000 Epochs (Norm, Off)



***Preds Normalized  
to Analytical Range***

*Raw output range is  
[24.0, 168.0]*

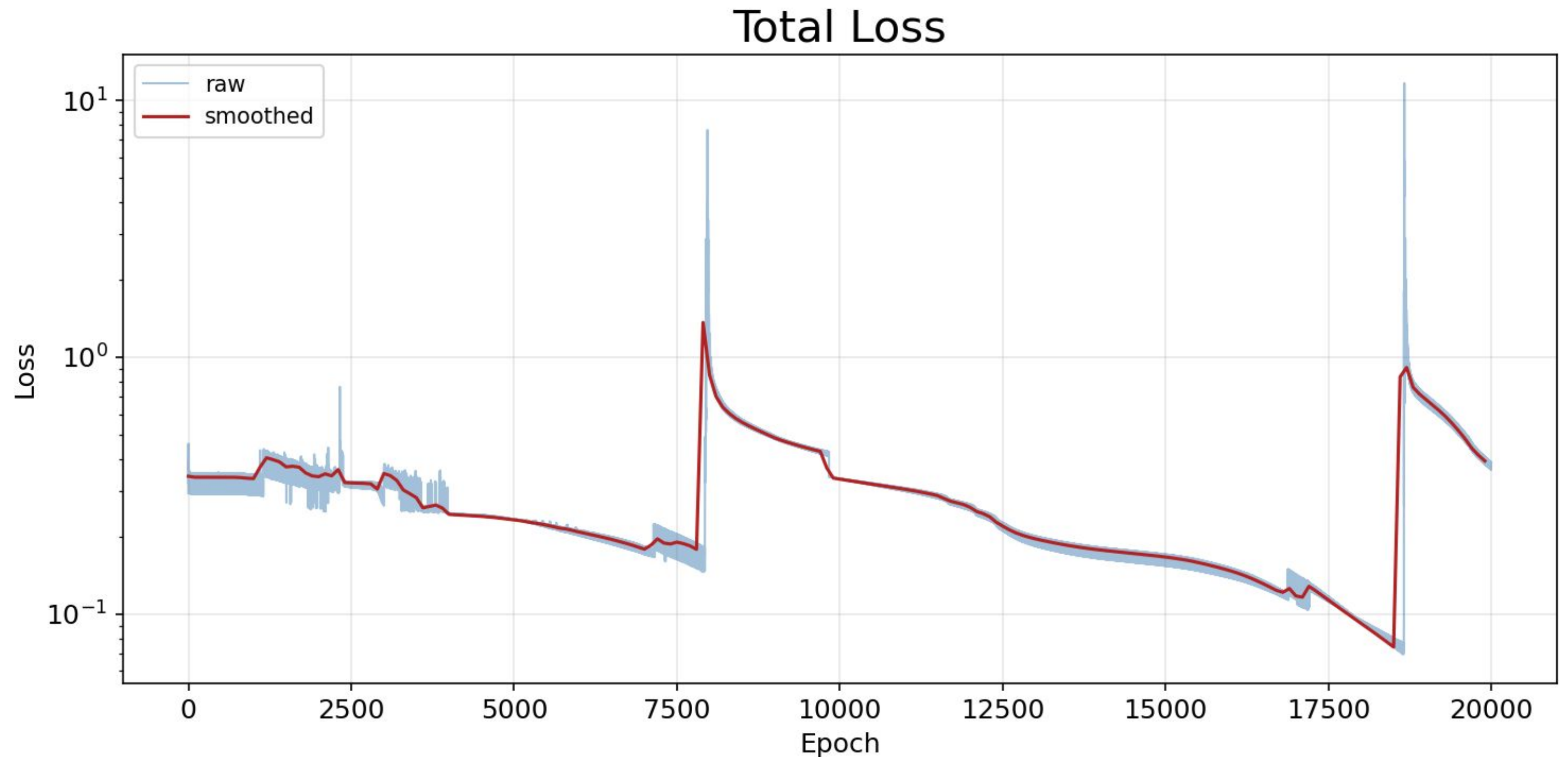
*Top is on offset grid,  
bottom is not*





## Observations

- 5 hidden layers, 128 neurons
- TanH hidden, final Softplus
- Lr start at  $1e-3$ , ends at  $4e-7$
- Converge at  $1e7$
- Model has moments where it will spikes, then refine
- Increment predictions minimum to minimum BC ambient temperature
- Adding per-part convergence and starting with all BCs turned on prevents slipping
- Residuals are very small so retain\_grad placement on tensors is essential
- Sampling grid size needs to balance gradient density with the resource requirements



# Future Work

## Model Validation

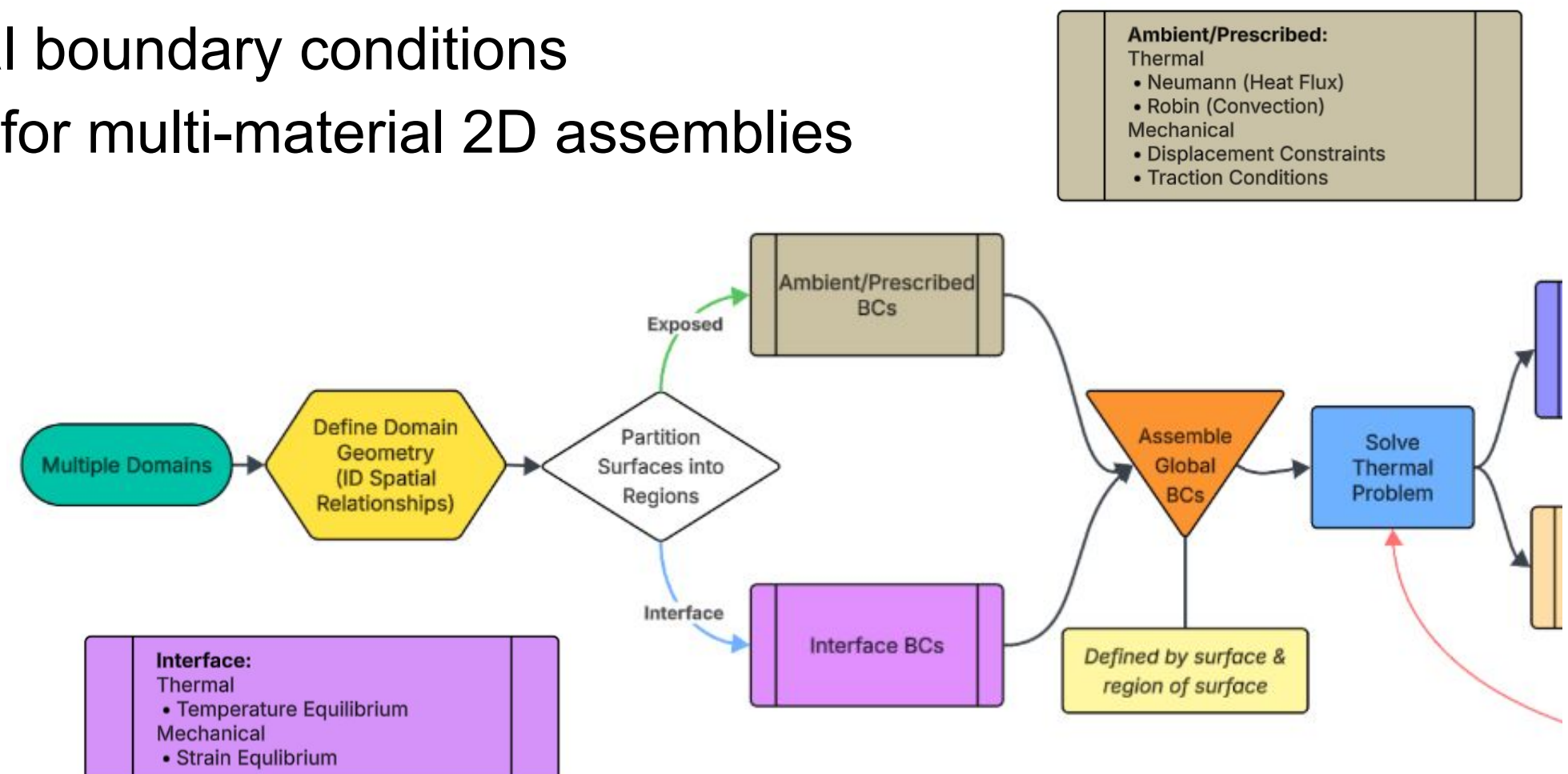
- Refine Thermal PINN performance (scaling, boundary conditions, hyperparameters)

## Model Development

- Develop the mechanical modeling PINN framework
- Implement flexible thermal and mechanical boundary conditions
- Incorporate interface boundary conditions for multi-material 2D assemblies

## Model Expansion

- Extend the framework to 3D domains



# Tools, Packages & Access

| Category                                 | Tool / Method                                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Machine Learning (ML) Approach           | Physics-Informed Neural Network (PINN)                                                                                      |
| Primary Package                          | PyTorch Autograd                                                                                                            |
| Integrated Development Environment (IDE) | PyCharm                                                                                                                     |
| AI Tools                                 | ChatGPT, Claude, Gemini<br><i>Assistance with translating theoretical concepts into code implementations and debugging.</i> |
| Code Repository                          | <a href="#">GitHub</a>                                                                                                      |



# References

- [1] MathWorks. (n.d.). *What are physics-informed neural networks (PINNs)?* MATLAB & Simulink. Retrieved June 18, 2026, from [MathWorks PINNs Overview](#)
- [2] Sharma, R., & Guo, Y. B. (2025). *Thermo-mechanical physics-informed deep learning for prediction of thermal stress evolution in laser metal deposition. Engineering Applications of Artificial Intelligence*, 157, 111554.  
<https://doi.org/10.1016/j.engappai.2025.111554>
- [3] Yu, X., Liu, Z., Li, H., Li, Y., Ai, X., Zeng, Z., Young, I. A., & Zhang, Z. (2025). *DeepOHeat-v1: Efficient operator learning for fast and trustworthy thermal simulation and optimization in 3D-IC design* (Version 2). arXiv.  
<https://doi.org/10.48550/arXiv.2504.03955>

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# ML Training

